CLINICAL RESEARCH & TECHNICAL AUDIT

aBCG Adult BCG Vaccine Effectiveness Study

Business Rules Implementation & Decision Logic Audit Report

System Architecture: Central Intake Wizard, Diagnostic Engines, Multi-Cohort Persistence & Excel Reporting

Document Control ID: AUDIT-BR-2026-V1
Audit Date: **29 October 2026**
Application Mode: **Production / Training System**
READ-ONLY STATIC VERIFICATION

TARGET AUDIENCE	INSPECTION SCOPE	CORE CODE LOCATION	ACTION REQUIRED
Head of Department (HOD) & PI	**Eligibility, Case/Control, EPTB, CXR, BCG**	`questions/views.py#L335-L1530`	**Review & Protocol Sign-Off**

1. EXECUTIVE IMPLEMENTATION SUMMARY

This technical audit was performed by line-by-line static inspection of the actual Python and template source code without altering any database records or logic. The investigation verified **22 discrete business rules** that govern participant screening, cohort placement, clinical disease classification, and surveillance analytics.

TOTAL DISCOVERED	FULLY IMPLEMENTED	PARTIALLY IMPLEMENTED	NOT FOUND	POTENTIAL PROBLEMS	PROTOCOL CONF. NEEDED
22 Rules	**17** Active in Code	**4** Mock / Generic	**1** "Pending BCG" choice	**6** Priority Conflicts	**3** Policy Decisions

Key Architectural Findings:

- **Unified Engine:** All participant intake and evaluation logic resides in `questions/views.py` within `def registration(request, pk=None)` (lines 335 to 1530).
- **Multi-Cohort Isolation:** Eligible study participants are stored in `Participant`; TPT exploratory participants in `TptIndividual`; and ineligibles in `IneligibleIndividual`.
- **Consent Refusal Hard-Stop:** If a participant declines informed consent (`consent == "No"`), data collection stops immediately in the frontend wizard, all clinical fields are blanked on the backend, and status is marked EXCLUDED.
- **High-Risk Priority Conflicts:** The sequential `if / elif` structure evaluates incomplete BCG details, TPT status, and 30-day investigation timeouts **before** the `Case` diagnostic criteria, creating risks of misclassifying true active TB cases.

2. MASTER CLASSIFICATION DECISION FLOWCHART (IMPLEMENTED CONTROL FLOW)

This flowchart reflects the **exact sequential order of checks** executed by the Python runtime in `questions/views.py` lines 1055 to 1105.



3. CLASSIFICATION PRIORITY & EXECUTION SEQUENCE

In Python, chained `if ... elif ... else` statements are evaluated strictly from top to bottom. The first condition that evaluates to `True` immediately sets the classification and **skips all remaining conditions**. The table below traces each step in order of execution:

Order	Gate / Condition	Output Status	Code Location & Syntax	Clinical Control Flow Analysis
1	**Study Eligibility Gate**	NOT ELIGIBLE	`views.py:1058` `if not eligible:`	First gate. If participant has no target HRG or was not eligible for BCG during the campaign, they are routed to `IneligibleIndividual`. Never reaches consent or testing.
2	**Consent Refusal Gate**	EXCLUDED	`views.py:1061` `elif consent_declined:`	Triggered if `consent == "No"`. Clears all diagnostic tests and BCG fields. Stored in `Participant` as an excluded record.
3	**BCG Verification Gate**	PENDING	`views.py:1076` `elif not bcg_complete:`	**Preempts Case Check:** If participant was vaccinated but date, batch number, or facility is omitted, status is set to Pending. If this participant is NAAT+, they become Pending rather than Case!
4	**TPT + BCG Cohort Gate**	TPT+BCG	`views.py:1080` `elif is_tpt_bcg:`	**Preempts Case Check:** If participant had prior TPT and BCG, routed to `TptIndividual`. If this participant develops active TB with positive NAAT, they are categorized as TPT+BCG instead of Case.
5	**Non-TB Abnormal CXR**	EXCLUDED	`views.py:1082` `elif cxr=="ABNORMAL_NON_TB" and mctb=="NEGATIVE":`	Excludes participants who have abnormal non-TB pulmonary findings on CXR when sputum NAAT is confirmed negative.
6	**Overdue Investigation Gate**	EXCLUDED	`views.py:1085` `elif (now - created_at).days > 30 and any_pending:`	**Critical Defect:** If record was created >30 days ago and *any* test is pending. If a patient is NAAT Positive (active TB!), but their CXR is still pending after 30 days, they are **Excluded** instead of being recorded as a **Case**!
7	**CASE Diagnostic Gate**	CASE	`views.py:1088` `elif mctb=="POSITIVE" or cxr=="SUGGESTIVE_OF_TB" or eptb=="POSITIVE":`	Matches confirmed microbiological TB, radiological TB on CXR, or positive extrapulmonary TB investigation.
8	**CONTROL Diagnostic Gate**	CONTROL	`views.py:1103` `elif cxr=="NORMAL" and mctb=="NEGATIVE" and eptb_neg:`	Requires normal CXR, negative NAAT, and negative/absent EPTB. Asymptomatic participants who do not undergo sputum testing cannot reach this branch.
9	**Default Fallback State**	PENDING	`views.py:1055, 1106` `else (default)`	Default state if none of 1–8 matched. Lines 1106–1128 populate an itemized reason naming the exact missing diagnostic investigations.

4. MASTER BUSINESS RULES IMPLEMENTATION MATRIX (ALL 22 RULES)

Rule Name	Status	Actual Conditions in Code	Assigned Output	File & Line	Audit Fi Implem Details
CASE	IMPLEMENTED	`mctb=="POSITIVE" || cxr=="SUGGESTIVE_OF_TB" || eptb_status=="POSITIVE"`	classification = "Case"	`questions/views.py#L1088-L1101`	Matches sputum N CXR sugg TB, or po EPTB finc
CONTROL	IMPLEMENTED	`cxr=="NORMAL" && mctb=="NEGATIVE" && (eptb_symptoms=="NO" || eptb_status=="NEGATIVE")`	classification = "Control"	`questions/views.py#L1103-L1104`	Requires sputum N Untested asympto stay Penc

Rule Name	Status	Actual Conditions in Code	Assigned Output	File & Line	Audit Fi Implem Details
PENDING	IMPLEMENTED	Default fallback state OR incomplete BCG OR missing test results	classification = "Pending"	questions/views.py#L1055, L1106-L1128	Populate itemized reason na missing M CXR, or E
EXCLUDED (Consent)	IMPLEMENTED	consent == "No"	classification = "Excluded"	questions/views.py#L1061-L1075	UI stops backend clinical fi saved to Participar
EXCLUDED (Non-TB CXR)	IMPLEMENTED	cxr == "ABNORMAL_NON_TB" && mctb == "NEGATIVE"	classification = "Excluded"	questions/views.py#L1082-L1084	Excludes pulmona findings NAAT is negative.
EXCLUDED (>30d Overdue)	IMPLEMENTED	(now - created_at).days > 30 && any_pending	classification = "Excluded"	questions/views.py#L1085-L1087	Evaluate Check 6; Case che (Potentia PRB-01).
Not Eligible	IMPLEMENTED	!has_hrg || !eligible_bcg_campaign	classification = "Not Eligible"	questions/views.py#L1058-L1060, L1429	Stored in Ineligible table; del Participar
Pending BCG	PARTIAL	bcg_status.startswith("Yes") but date, batch, or facility missing	classification = "Pending"	questions/views.py#L1076-L1078	Stored as "Pending separate BCG" stat models.
TPT + BCG Cohort	IMPLEMENTED	has_hrg && eligible_bcg && tpt=="Yes" && bcg=="Yes"	classification = "TPT+BCG"	questions/views.py#L1053, L1080	Stored in TptIndivi table. Eva before Ca (PRB-02).
HRG: Elderly (>=60)	IMPLEMENTED	age >= 60 || "Individuals aged 60 years or above" in risk_factors	is_elderly = True	questions/views.py#L564-L569	Auto-der age input factors ch
HRG: Malnourished	IMPLEMENTED	BMI < 18 || "Undernourished / Malnourished" in risk_factors	is_malnourished = True	questions/views.py#L570-L575	Auto-der height/w formula (checkbox
HRG: Close Contact	IMPLEMENTED	Contact of index TB case enrolled in Nikshay from 1 Jan 2021	is_contact = True	questions/views.py#L576-L580	Matched master ri list and B criteria.
HRG: Diabetes	IMPLEMENTED	Self-reported diabetes during BCG campaign period	is_diabetes = True	questions/views.py#L581-L585	Matched master ri list and B criteria.
HRG: Past TB	IMPLEMENTED	Episode of TB in past 5 years (2019–2024)	is_past_tb = True	questions/views.py#L586-L590	Matched master ri list and B criteria.
HRG: Smoker	IMPLEMENTED	History of smoking tobacco (Current / Past User)	is_smoker = True	questions/views.py#L591-L596	Matched master ri list and B criteria.

Rule Name	Status	Actual Conditions in Code	Assigned Output	File & Line	Audit Fi Impleme Details
NAAT / MCTB Parsing	IMPLEMENTED	String inspection of final_interpretation & naat_status	**mctb = POSITIVE | NEGATIVE | PENDING**	`questions/views.py#L965-L980`	MTB Det Rif Resist Positive; Detected Negative
Chest X-Ray Status	IMPLEMENTED	Suggestive of TB / Abnormal Non-TB / Normal / Not Done	**cxr = SUGGESTIVE | ABNORMAL | NORMAL**	`questions/views.py#L983-L996`	Asympto with cxr_done are auto- to NORM
EPTB 11-Site Engine	IMPLEMENTED	11 site questions + dynamic investigation performed/findings	**eptb_status = POSITIVE | NEGATIVE | PENDING**	`questions/views.py#L902-L932, L999-L1039`	Positive investiga Case; syn without t Pending.
Enrollment Date Preserv.	IMPLEMENTED	Existing record -> target_pk -> hidden POST -> today fallback	**date_enroll preserved; updated_at saved**	`questions/views.py#L1400-L1428`	Verified v passing automate regressio
Location Anti-Corruption	IMPLEMENTED	Priority: participant -> form -> nurse profile -> session	**Never overwrites with 'GLOBAL'**	`questions/locations_data.py#L354-L410`	Protects participa coordina admin us profiles.
Dashboard KPI Counts	IMPLEMENTED	`filter(classification="Case")`, `Control`	**confirmed_cases, study_controls**	`questions/views.py#L3686-L3729`	Total Enr strictly de Cases + C
Adjudication Workflow	PARTIAL	`filter(classification="Adjudication Pending")`	Never assigned by intake engine	`questions/views.py#L4424, L4750`	Adjudica queue al back to c mock dat

5. DIAGNOSTIC INVESTIGATION COMBINATIONS & ASSIGNED OUTCOMES

The table below traces how the eligibility engine in `views.py#L1082-L1105` classifies every combination of NAAT, Chest X-Ray, and EPTB findings:

Sputum NAAT (MCTB)	Chest X-Ray (CXR)	EPTB Status	Assigned Status	Exact Classification Reason Generated	Source Line
POSITIVE	SUGGESTIVE_OF_TB	Any	CASE	"MCTB Positive and CXR suggestive of TB"	views.py:1090
POSITIVE	NORMAL	Any	CASE	"MCTB Positive and Normal CXR"	views.py:1092
POSITIVE	ABNORMAL_NON_TB	Any	CASE	"MCTB Positive"	views.py:1094
POSITIVE	PENDING / NOT_DONE	Any	CASE	"MCTB Positive"	views.py:1094
NEGATIVE	**SUGGESTIVE_OF_TB**	Any	CASE	"CXR suggestive of TB with MCTB Negative"	views.py:1096
NEGATIVE	ABNORMAL_NON_TB	Any	EXCLUDED	"Abnormal CXR not suggestive of TB with NAAT Negative"	views.py:1083
NEGATIVE	**NORMAL**	NEGATIVE / NONE	CONTROL	"Normal CXR, Negative NAAT, and Negative EPTB status"	views.py:1103
NEGATIVE	NORMAL	**POSITIVE**	CASE	"EPTB Investigation Positive"	views.py:1101
NEGATIVE	NORMAL	PENDING	PENDING	"EPTB is pending because investigation results are not recorded"	views.py:1122
PENDING	**SUGGESTIVE_OF_TB**	Any	CASE	"Chest X-Ray suggestive of TB"	views.py:1098
PENDING	ABNORMAL_NON_TB	Any	PENDING	"NAAT (MCTB) is pending..."	views.py:1108
PENDING	NORMAL	NEGATIVE / NONE	PENDING	"NAAT (MCTB) is pending because clinical testing not undergone"	views.py:1108
PENDING	PENDING / NOT_DONE	Any	PENDING	"NAAT (MCTB) is pending...; Chest X-Ray has not been performed"	views.py:1114

6. POTENTIAL BUSINESS RULE PROBLEMS & LOGIC INCONSISTENCIES (6 FINDINGS)

PRB-01 · HIGH SEVERITY `questions/views.py#L1085-L1089`

Overdue Pending Rule Excludes Confirmed Microbiological TB Cases

Actual Behavior: Check 6 evaluates `(now - created_at).days > 30 and any_pending` *before* Check 7 (`Case`). If a participant is NAAT Positive (microbiologically confirmed TB), but their secondary Chest X-Ray was left pending or delayed for >30 days, Check 6 matches first and classifies them as **`Excluded`** ("Pending investigation beyond one month").

```
Example Input: mctb="POSITIVE", cxr="PENDING", created_at=35 days ago Actual Output: classification = "Excluded" (EXPECTED: "Case")
```

Recommended Fix: Move Check 7 (`Case`) above Check 6. Confirmed active disease must never be discarded due to pending secondary investigations.

PRB-02 · HIGH SEVERITY

questions/views.py#L1080-L1089

TPT+BCG Cohort Check Preempts Active TB Case Detection

Actual Behavior: Check 4 evaluates `is_tpt_bcg` *before* Check 7 (`Case`). If a participant who underwent TPT and received BCG develops active pulmonary TB with positive NAAT, they are saved as `classification = "TPT+BCG"` in the `TptIndividual` table rather than being recorded as an active TB Case.

```
Example Input: tpt_undergone="Yes", bcg_status="Yes", mctb="POSITIVE" Actual Output: classification = "TPT+BCG" in TptIndividual table (EXPECTED: "Case")
```

Recommended Fix: Protocol confirmation required. Active TB breakthrough cases in TPT recipients should either be classified as Cases or tracked under a dual-reporting flag.

PRB-03 · MEDIUM SEVERITY

questions/views.py#L1076-L1079

Incomplete BCG Details Suppresses Confirmed TB Case Diagnosis

Actual Behavior: Check 3 tests `not bcg_complete`. If a nurse records `bcg_status="Yes"` but leaves the batch number or facility blank, `bcg_complete` evaluates to `False`. Because Check 3 runs before Check 7, a participant with a positive sputum NAAT is classified as **`Pending`** ("One or more required BCG verification or vaccine details are incomplete") instead of a confirmed **`Case`**.

```
Example Input: mctb="POSITIVE", bcg_status="Yes", bcg_batch_number="" Actual Output: classification = "Pending" (EXPECTED: "Case")
```

Recommended Fix: Clinical disease status (Case) should be evaluated first; BCG record completeness should be tracked as a verification sub-status.

PRB-04 · MEDIUM SEVERITY

questions/views.py#L977, L1103

Untested Asymptomatic Participants Can Never Qualify as Controls

Actual Behavior: For asymptomatic participants who do not undergo sputum testing (`undergone_testing == "No"`), line 977 assigns `mctb = "PENDING"`. Check 8 requires `mctb == "NEGATIVE"` to qualify as a Control. As a result, asymptomatic participants without sputum tests remain indefinitely in **`Pending`** status.

```
Example Input: is_asymptomatic=True, cxr="NORMAL", undergone_testing="No" Actual Output: classification = "Pending"
```

Protocol Decision Required: Does the study protocol require asymptomatic community controls to submit sputum for NAAT testing, or does asymptomatic status with normal/absent CXR qualify as a Control?

PRB-05 · LOW SEVERITY

questions/views.py#L4424, L4750

Orphaned "Adjudication Pending" & "Needs Correction" Dashboard States

Actual Behavior: The admin adjudication dashboard filters by `classification="Adjudication Pending"` and `classification="Needs Correction"`, but neither state is part of `Participant.CLASSIFICATION_CHOICES` or assigned anywhere in the application. As a result, the adjudication queue always falls back to central mock data.

Recommended Fix: Either implement the supervisor adjudication transition workflow or decouple adjudication into a separate database field rather than overloading `classification`.

PRB-06 · LOW SEVERITY

questions/views.py#L1085

Duration Calculation Uses Record Creation Date Instead of Clinical Date

Actual Behavior: The 30-day pending exclusion rule computes `(timezone.now() - participant.created_at).days > 30`. If an existing record was imported or created weeks before actual clinical intake, the duration clock starts prematurely from database insert timestamp rather than `participant.date_enroll`.

Recommended Fix: Use `participant.date_enroll` as the clinical reference date for pending duration calculations.

7. PROTOCOL CONFIRMATION & HOD ACTION ITEMS

Please review the following four core clinical decision rules with the Head of Department (HOD) and Protocol Committee. Mark the approved decision for each item to guide technical adjustments:

Decision 1: Active TB Breakthrough in TPT + BCG Recipients (Ref: PRB-02) PRIORITY 1

When a participant who completed TPT and received BCG develops microbiologically confirmed active TB (NAAT Positive):

- ☐ **Option A (Recommended):** Classify as **CASE** in primary cohort, with TPT history recorded as an exploratory covariate.
- ☐ **Option B (Current Code):** Retain classification as **TPT+BCG** in the separate TPT individual registry.

Decision 2: Control Group NAAT Sputum Testing Requirement (Ref: PRB-04) PRIORITY 2

For asymptomatic high-risk individuals with normal or absent Chest X-Rays:

- ☐ **Option A (Recommended):** Asymptomatic status with Normal or Not Done CXR qualifies as a **CONTROL** without requiring sputum NAAT.
- ☐ **Option B (Current Code):** Sputum NAAT is mandatory; untested asymptomatics must remain in **PENDING** status.

Decision 3: Overdue Pending Investigation Exclusion Window (Ref: PRB-01 & PRB-06) PRIORITY 3

When diagnostic investigation results are delayed:

- ☐ **Option A (Recommended):** Move Case check **ABOVE** duration check (a positive NAAT or CXR is never excluded); calculate timeout from `date_enroll`.
- ☐ **Option B:** Increase pending exclusion window from 30 calendar days to 60 or 90 calendar days.
- ☐ **Option C:** Disable automatic time-based exclusion; flag overdue tests for supervisor review without changing status.

Decision 4: Incomplete BCG Details vs Active TB Diagnosis (Ref: PRB-03) PRIORITY 4

When an active TB patient has positive sputum NAAT, but their BCG batch number or facility name is not yet retrieved:

- ☐ **Option A (Recommended):** Classify as **CASE** immediately, with BCG completeness tracked as an operational verification flag.
- ☐ **Option B (Current Code):** Retain participant in **PENDING** status until BCG batch details are entered.

FORMAL PROTOCOL REVIEW & APPROVAL SIGN-OFF

HEAD OF DEPARTMENT (HOD)	PRINCIPAL INVESTIGATOR (PI)	STUDY BIOSTATISTICIAN
Signature: _____	Signature: _____	Signature: _____
Date: ___ / ___ / 2026	Date: ___ / ___ / 2026	Date: ___ / ___ / 2026